

Qualification Pack



AI Data Quality Analyst

QP Code: SSC/Q8101

Version: 4.0

NSQF Level: 5

IT-ITeS SSC NASSCOM || NASSCOM Plot No - 7, 8, 9 & 10, 3rd Floor, Sector 126
Noida Uttar Pradesh - 201303 || email:sscstandards@nasscom.in

Qualification Pack

Contents

SSC/Q8101: AI Data Quality Analyst	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
SSC/N8101: Extract or Import data as per requirement	5
SSC/N8102: Process the data	9
SSC/N8163: Ensure organizational quality standards are achieved regarding data	13
SSC/N8164: Application of new and emerging tools	18
SSC/N8165: Implement and Enforce Data Governance Policies	23
DGT/VSQ/N0102: Employability Skills (60 Hours)	28
Assessment Guidelines and Weightage	35
<i>Assessment Guidelines</i>	35
<i>Assessment Weightage</i>	35
Acronyms	37
Glossary	38

Qualification Pack

SSC/Q8101: AI Data Quality Analyst

Brief Job Description

Data Quality analyst qualification is designed to equip individuals with the knowledge and practical expertise to manage, ensure, and improve the quality, security, and integrity of organisational data. This qualification covers the end-to-end data management lifecycle including sourcing, validation, transformation, monitoring, security, and governance using industry best practices, tools, and frameworks. It prepares learners to establish data standards, implement automated workflows, conduct audits, and foster a culture of data ownership, compliance, and continuous improvement across the organisation.

Personal Attributes

A data quality analyst needs to have good analytical skills, attention to detail and problem solving ability. S/he needs to have good communication skills to work with stakeholders across multiple teams such as data science, product development, etc.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8101: Extract or Import data as per requirement](#)
2. [SSC/N8102: Process the data](#)
3. [SSC/N8163: Ensure organizational quality standards are achieved regarding data](#)
4. [SSC/N8164: Application of new and emerging tools](#)
5. [SSC/N8165: Implement and Enforce Data Governance Policies](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
Country	India
NSQF Level	5

Qualification Pack

Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/252.NIL
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (UG/diploma with courses related to Engg./ Science) with NA of experience OR Completed 3-year diploma (after 10th) (UG/diploma with courses related to Engg./ Science) with 1.5 years of experience in Relevant Field- AI/Big Data/Computer Science/IT .** Relevant Field- AI/Big Data/Computer Science/IT The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience in Relevant Field- AI/Big Data/Computer Science/IT .** Relevant Field- AI/Big Data/Computer Science/IT The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	22 Years
Last Reviewed On	NA
Next Review Date	12/03/2029
NSQC Approval Date	12/03/2026
Version	4.0
Reference code on NQR	QG-05-IT-06001-2026-V2-NASSCOM
NQR Version	2

Qualification Pack

SSC/N8101: Extract or Import data as per requirement

Description

This unit is about using a variety of techniques to import data into datasets or data frames.

Scope

The scope covers the following :

- Identify Data Requirements, Develop a Data Collection Strategy, Set Up Data Import Mechanisms

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Determine what data is needed based on business goals and analytical needs
- PC2.** Identify potential internal and external sources such as databases, APIs, web services, spreadsheets, and third-party vendors.
- PC3.** Evaluate the accessibility, reliability, and quality of each source
- PC4.** Decide how often data needs to be collected (real-time, hourly, daily, etc.).
- PC5.** Choose appropriate methods for data retrieval, including API calls, data feeds, and file transfers.
- PC6.** Opt for ETL (Extract, Transform, Load) tools or data integration platforms suited to the business needs
- PC7.** Create workflows for automating data extraction, transformation, and loading into target systems.
- PC8.** Establish secure and reliable connections to each data source.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding data governance policies procedures and guidelines and knowing how to access information from data sources
- KU2.** Familiarity with organizational policies for sharing and documenting data governance processes and involving the right stakeholders
- KU3.** Knowledge of different database types data modeling SQL skills and database management system platforms
- KU4.** Skills in ETL processes performance tuning access control backup and recovery and database monitoring and maintenance
- KU5.** Insights into scalability practices data integrity documentation collaboration and continuous learning for effective database management

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** make decisions on suitable course of action
- GS2.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Determine what data is needed based on business goals and analytical needs	3	6	-	2
PC2. Identify potential internal and external sources such as databases, APIs, web services, spreadsheets, and third-party vendors.	3	6	-	2
PC3. Evaluate the accessibility, reliability, and quality of each source	3	6	-	2
PC4. Decide how often data needs to be collected (real-time, hourly, daily, etc.).	4	6	-	2
PC5. Choose appropriate methods for data retrieval, including API calls, data feeds, and file transfers.	4	6	-	3
PC6. Opt for ETL (Extract, Transform, Load) tools or data integration platforms suited to the business needs	4	6	-	3
PC7. Create workflows for automating data extraction, transformation, and loading into target systems.	4	6	-	3
PC8. Establish secure and reliable connections to each data source.	5	8	-	3
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8101
NOS Name	Extract or Import data as per requirement
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

SSC/N8102: Process the data

Description

This unit is about using a variety of techniques to process data i.e. clean and transform the data

Scope

The scope covers the following :

- Data Preparation & standardization, Data Loading and Integration, Monitoring and Maintenance

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Apply transformations to clean and harmonize data into a uniform format before import
- PC2.** Define data schemas and structures that align with organizational databases.
- PC3.** Set basic validation checks during import to ensure data quality (e.g., format checks, range checks).
- PC4.** Transfer processed data into data warehouses, databases, or other storage solutions.
- PC5.** For real-time needs, implement streaming data solutions where applicable
- PC6.** Ensure new data is merged seamlessly with existing datasets to provide a consistent view.
- PC7.** Set up alerts and monitoring to catch issues in the data import process quickly.
- PC8.** Regularly assess import processes to identify bottlenecks or errors, and resolve them promptly.
- PC9.** Maintain comprehensive documentation of data collection strategies, source configurations, and process workflows for easy reference and modification.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding of organizational guidelines for data accuracy, consistency, security, and compliance when processing and importing data.
- KU2.** Knowledge of internal data warehouses, databases, and storage systems where processed data must be integrated or loaded.
- KU3.** Familiarity with business-defined schemas, master data definitions, and data relationships to ensure correct data harmonization and merging.
- KU4.** Awareness of how data engineers, analysts, IT teams, and data stewards collaborate in data transformation, validation, monitoring, and documentation.
- KU5.** Understanding organizational procedures for updating import workflows, handling data pipeline failures, version control, and documentation standards.
- KU6.** Knowledge of methods like normalization, standardization, deduplication, handling missing values, and unifying data formats before import.

Qualification Pack

- KU7.** Understanding relational schemas, star/snowflake models, primary/foreign keys, and structural rules required for aligning data with organizational databases.
- KU8.** Knowledge of automated validation techniques such as format checks, range checks, schema validation, and error-handling mechanisms during import.
- KU9.** Familiarity with tools such as Airflow, Talend, Informatica, AWS Glue, Azure Data Factory, Kafka/streaming technologies, and loading techniques for warehouses and databases.
- KU10.** Understanding logging systems, alerting tools, process monitoring dashboards, troubleshooting methods, and optimization techniques for efficient data imports.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to identify data issues, bottlenecks, inconsistencies, and devise efficient solutions during processing and import.
- GS2.** Skill in ensuring that transformations, schema mappings, validations, and merges are performed precisely to maintain data integrity.
- GS3.** Ability to clearly document workflows, configurations, changes, and coordinate with teams to ensure smooth data pipeline operations.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Apply transformations to clean and harmonize data into a uniform format before import	1	2	-	2
PC2. Define data schemas and structures that align with organizational databases.	2	3	-	2
PC3. Set basic validation checks during import to ensure data quality (e.g., format checks, range checks).	3	5	-	2
PC4. Transfer processed data into data warehouses, databases, or other storage solutions.	4	7	-	3
PC5. For real-time needs, implement streaming data solutions where applicable	5	8	-	2
PC6. Ensure new data is merged seamlessly with existing datasets to provide a consistent view.	5	8	-	2
PC7. Set up alerts and monitoring to catch issues in the data import process quickly.	5	8	-	2
PC8. Regularly assess import processes to identify bottlenecks or errors, and resolve them promptly.	4	7	-	3
PC9. Maintain comprehensive documentation of data collection strategies, source configurations, and process workflows for easy reference and modification.	1	2	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8102
NOS Name	Process the data
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	5
Version	3.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

SSC/N8163: Ensure organizational quality standards are achieved regarding data

Description

This unit is about ensuring organizational quality standards regarding data

Scope

The scope covers the following :

- Define Data Quality Standards, Develop Quality Assurance Processes, Establish Monitoring Mechanisms, Conduct Regular Audits and Reviews, Prioritize Continuous Improvement, Foster Collaborative Culture, Documentation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Establish clear criteria for data quality dimensions like accuracy, completeness, consistency, timeliness, validity, and relevancy
- PC2.** Determine acceptable thresholds and benchmarks for each quality dimension, tailored to the organization's specific needs and objectives.
- PC3.** Create processes and controls to regularly assess data against the defined standards. This can include automated scripts, manual reviews, and cross-verification with trusted sources.
- PC4.** Encode business logic and rules to validate incoming data, ensuring it meets necessary criteria before entering critical systems
- PC5.** Use data quality tools to automate monitoring and alerts for data anomalies, deviations, or breaches in quality standards.
- PC6.** Implement dashboards that provide visual indicators of data quality metrics in real time, enabling quick detection and response to issues.
- PC7.** Schedule regular audits to thoroughly review datasets and validate that they comply with quality standards.
- PC8.** Perform gap analysis to identify discrepancies between current data quality and desired standards, providing a foundation for improvements.
- PC9.** Establish feedback loops with end-users and stakeholders to gather insights on data quality issues and suggestions for enhancement.
- PC10.** Use findings from audits and monitoring to refine quality assurance processes continuously, adapting to the evolving needs of the business.
- PC11.** Work closely with IT, data science, and business teams to ensure a unified approach to data quality.
- PC12.** Provide training for staff on data quality principles and their importance, fostering a culture of quality throughout the organization.
- PC13.** Maintain detailed records of data quality standards, processes, checks, and incidents to enhance transparency and guide future initiatives.

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding of organizational rules, regulatory requirements, and governance policies related to data accuracy, privacy, integrity, and audit readiness.
- KU2.** Awareness of the organizations strategic goals, KPIs, and how data quality impacts decision-making, operations, and customer outcomes.
- KU3.** Knowledge of designated data owners, stewards, cross-functional teams, and escalation paths responsible for maintaining and enforcing data quality standards.
- KU4.** Familiarity with the databases, data pipelines, applications, data warehouses, and analytics platforms where data quality checks must be applied.
- KU5.** Understanding of internal procedures for documenting standards, updating business rules, implementing corrective measures, and conducting periodic reviews.
- KU6.** Knowledge of accuracy, completeness, consistency, timeliness, validity, and relevancy, and how to define thresholds and quantify these metrics.
- KU7.** Understanding how to design, encode, and apply validation rules, business logic, cross-verification checks, and data profiling methods.
- KU8.** Familiarity with tools such as Talend DQ, Informatica DQ, Ataccama, Great Expectations, and monitoring systems for anomaly detection, scoring, and alerts.
- KU9.** Knowledge of BI tools (Power BI, Tableau, Looker) used to build visual indicators for real-time quality monitoring and issue detection.
- KU10.** Understanding of how to perform data audits, conduct gap analyses, monitor pipelines, review logs, track incidents, and refine quality processes.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to assess data discrepancies, identify root causes, evaluate quality gaps, and apply logical problem-solving for continuous improvement.
- GS2.** Capability to work with IT, business users, data science teams, and stakeholders, while clearly communicating quality issues, standards, and required corrective actions.
- GS3.** Skill in creating detailed records of standards, checks, processes, and incidents while maintaining accuracy and precision across all quality activities.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Establish clear criteria for data quality dimensions like accuracy, completeness, consistency, timeliness, validity, and relevancy	2	3	-	1
PC2. Determine acceptable thresholds and benchmarks for each quality dimension, tailored to the organization's specific needs and objectives.	2	3	-	1
PC3. Create processes and controls to regularly assess data against the defined standards. This can include automated scripts, manual reviews, and cross-verification with trusted sources.	2	4	-	1
PC4. Encode business logic and rules to validate incoming data, ensuring it meets necessary criteria before entering critical systems	2	4	-	1
PC5. Use data quality tools to automate monitoring and alerts for data anomalies, deviations, or breaches in quality standards.	2	4	-	1
PC6. Implement dashboards that provide visual indicators of data quality metrics in real time, enabling quick detection and response to issues.	2	4	-	1
PC7. Schedule regular audits to thoroughly review datasets and validate that they comply with quality standards.	3	4	-	2
PC8. Perform gap analysis to identify discrepancies between current data quality and desired standards, providing a foundation for improvements.	3	4	-	2
PC9. Establish feedback loops with end-users and stakeholders to gather insights on data quality issues and suggestions for enhancement.	3	4	-	2
PC10. Use findings from audits and monitoring to refine quality assurance processes continuously, adapting to the evolving needs of the business.	3	4	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Work closely with IT, data science, and business teams to ensure a unified approach to data quality.	2	4	-	2
PC12. Provide training for staff on data quality principles and their importance, fostering a culture of quality throughout the organization.	2	4	-	2
PC13. Maintain detailed records of data quality standards, processes, checks, and incidents to enhance transparency and guide future initiatives.	2	4	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8163
NOS Name	Ensure organizational quality standards are achieved regarding data
Sector	IT-ITeS
Sub-Sector	
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

SSC/N8164: Application of new and emerging tools

Description

This NOS about the application of new and emerging tools in AI

Scope

The scope covers the following :

- Research and Identify Emerging Tools and Technologies, Evaluate and Select Appropriate Tools, Implement and Integrate New Tools, Automate Data Quality Processes Using Advanced Tools, Automate Data Quality Processes Using Advanced Tools, Train and Support Team Members, Monitor the Performance and Effectiveness of Tools, Document Tool Implementation and Usage, Facilitate Continuous Improvement and Innovation, Collaborate with Cross-Functional Teams, Document Tool Implementation and Usage

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Stay informed of the latest trends and developments in data quality tools, software, and technologies
- PC2.** Analyze emerging tools for their applicability to the organization's data quality needs
- PC3.** Conduct pilot testing to determine tool effectiveness and integration capabilities
- PC4.** Make recommendations based on cost, efficiency, scalability, and user-friendliness
- PC5.** Work with IT teams to implement selected tools, ensuring smooth integration with existing data architectures.
- PC6.** Develop custom scripts or configurations if needed to enhance tool performance.
- PC7.** Use AI and machine learning technologies to automate routine data quality tasks, such as data cleansing and anomaly detection.
- PC8.** Set up automated workflows and alerts to monitor data quality in real-time.
- PC9.** Provide training sessions and resources to team members on using new tools effectively
- PC10.** Develop user guides and documentation to assist in tool usage and troubleshooting
- PC11.** Regularly assess the performance of implemented tools through key performance indicators (KPIs) and metrics
- PC12.** Adjust configurations and processes based on feedback and ongoing evaluation results.
- PC13.** Encourage a culture of experimentation and innovation within the data quality team.
- PC14.** Collect feedback from users to identify opportunities for enhancement and tool optimization.
- PC15.** Work closely with data scientists, IT specialists, and business users to align tool usage with organizational goals.
- PC16.** Ensure that new tool implementations support wider data-driven initiatives within the organization.
- PC17.** Maintain detailed records of tool evaluations, implementation processes, and user feedback.
- PC18.** Document best practices and lessons learned to support future tool adoptions and upgrades.

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the organizations long-term data strategy, governance models, security standards, and approved technology stack to ensure new tools align with enterprise policies.
- KU2.** Awareness of business goals, operational needs, and data-driven initiatives so that tool selection and implementation support measurable improvements in data quality.
- KU3.** Knowledge of how data teams, IT, data scientists, and business stakeholders collaborate for evaluating tools, conducting pilots, gathering feedback, and integrating new solutions.
- KU4.** Familiarity with internal approval workflows, cost-benefit assessment procedures, vendor management, and procurement policies for acquiring or upgrading tools.
- KU5.** Understanding of current data architectures, data pipelines, metadata systems, and platforms to ensure smooth integration and compatibility during tool deployment.
- KU6.** Knowledge of industry-leading platforms (e.g., Informatica DQ, Talend DQ, Ataccama, Great Expectations, Monte Carlo) and awareness of new advances in AI/ML-based quality tools.
- KU7.** Understanding how to assess tools for scalability, performance, usability, integration capability, cost efficiency, and ability to meet organizational quality requirements.
- KU8.** Ability to create custom scripts (Python, SQL, shell) and configure workflows, rules engines, and automated alerts for cleansing, anomaly detection, and continuous monitoring.
- KU9.** Knowledge of ML techniques used in automated data quality tasks pattern recognition, anomaly detection, classification, entity matching, and intelligent data cleansing.
- KU10.** Familiarity with KPIs, dashboards, tool performance metrics, logging systems, troubleshooting, and continuous improvement frameworks for evaluating tool effectiveness.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to evaluate new technologies, identify improvement opportunities, conduct pilot assessments, and innovate with AI/ML-driven quality solutions.
- GS2.** Skill in working with IT teams, business users, data scientists, and vendors, while providing clear documentation, guidance, training, and tool usage support.
- GS3.** Ability to troubleshoot issues, optimize configurations, adjust processes based on feedback, and adapt to fast-changing technologies and organizational needs.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Stay informed of the latest trends and developments in data quality tools, software, and technologies	1	2	-	1
PC2. Analyze emerging tools for their applicability to the organization's data quality needs	1	2	-	1
PC3. Conduct pilot testing to determine tool effectiveness and integration capabilities	1	2	-	1
PC4. Make recommendations based on cost, efficiency, scalability, and user-friendliness	1	2	-	1
PC5. Work with IT teams to implement selected tools, ensuring smooth integration with existing data architectures.	1	3	-	1
PC6. Develop custom scripts or configurations if needed to enhance tool performance.	1	3	-	1
PC7. Use AI and machine learning technologies to automate routine data quality tasks, such as data cleansing and anomaly detection.	2	3	-	1
PC8. Set up automated workflows and alerts to monitor data quality in real-time.	2	3	-	1
PC9. Provide training sessions and resources to team members on using new tools effectively	2	3	-	1
PC10. Develop user guides and documentation to assist in tool usage and troubleshooting	2	3	-	1
PC11. Regularly assess the performance of implemented tools through key performance indicators (KPIs) and metrics	2	3	-	1
PC12. Adjust configurations and processes based on feedback and ongoing evaluation results.	2	3	-	1
PC13. Encourage a culture of experimentation and innovation within the data quality team.	2	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Collect feedback from users to identify opportunities for enhancement and tool optimization.	2	3	-	1
PC15. Work closely with data scientists, IT specialists, and business users to align tool usage with organizational goals.	2	3	-	1
PC16. Ensure that new tool implementations support wider data-driven initiatives within the organization.	2	3	-	1
PC17. Maintain detailed records of tool evaluations, implementation processes, and user feedback.	2	3	-	2
PC18. Document best practices and lessons learned to support future tool adoptions and upgrades.	2	3	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8164
NOS Name	Application of new and emerging tools
Sector	IT-ITeS
Sub-Sector	
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQF Clearance Date	12/03/2026

Qualification Pack

SSC/N8165: Implement and Enforce Data Governance Policies

Description

The NOS is about implementing & enforcing the data governance policies

Scope

The scope covers the following :

- Implement and Enforce Data Governance Policies

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Develop Data Governance Framework and Policies
- PC2.** Communicate and Advocate Data Governance Policies
- PC3.** Define and enforce data standards including data definitions, classifications, quality benchmarks, and security protocols.
- PC4.** Establish metadata management practices to track data lineage, sources, and usage.
- PC5.** Employ tools and processes to monitor adherence to governance policies and standards
- PC6.** Conduct periodic reviews, audits, and risk assessments to detect non-compliance or data breaches.
- PC7.** Implement policies to safeguard sensitive and critical data, ensuring compliance with relevant regulations (e.g., GDPR, HIPAA).
- PC8.** Coordinate with IT and legal teams to enforce data access controls, encryption, and anonymization techniques.
- PC9.** Deploy and manage tools that facilitate policy enforcement, data cataloging, and data lineage tracking.
- PC10.** Leverage automation solutions for policy enforcement and monitoring
- PC11.** Establish procedures for handling data quality issues, policy breaches, or non-compliance incidents.
- PC12.** Lead corrective actions and communicate findings and resolutions to stakeholders.
- PC13.** Foster a Culture of Data Governance and Ownership
- PC14.** Maintain Documentation and Records

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the legal and regulatory environment (e.g., GDPR, HIPAA, industry-specific standards) that shapes organizational data governance policies and risk mitigation strategies.

Qualification Pack

- KU2.** Awareness of roles such as data owners, stewards, custodians, and governance committees, including escalation pathways for policy breaches, data quality issues, and compliance violations.
- KU3.** Understanding how governance supports business goals, analytical functions, digital transformation, and data-driven decision-making across departments.
- KU4.** Familiarity with the organizations databases, data cataloging systems, metadata repositories, access control mechanisms, and security frameworks used to implement governance controls.
- KU5.** Knowledge of organizational processes for introducing new policies, communicating governance updates, onboarding teams, and maintaining documentation.
- KU6.** Knowledge of frameworks like DAMA-DMBOK, ISO 8000, DCAM, and best-practice principles for defining data policies, classifications, data quality rules, and stewardship models.
- KU7.** Understanding of tools such as Collibra, Alation, Informatica EDC, Apache Atlas, and methodologies for capturing lineage, sources, transformations, and usage patterns.
- KU8.** Knowledge of encryption, anonymization, masking, RBAC, IAM systems, and security protocols required for safeguarding sensitive and critical data.
- KU9.** Familiarity with tools that support policy enforcement, cataloging, audit logging, automated monitoring, compliance alerts, and risk assessments.
- KU10.** Technical understanding of how to conduct audits, manage policy violations, perform root-cause analyses, document incidents, and drive corrective actions.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to advocate governance principles, educate stakeholders, lead corrective initiatives, and promote a culture of ownership and compliance.
- GS2.** Skill in identifying governance gaps, assessing risks, performing audits, and evaluating compliance with policies and data standards.
- GS3.** Ability to maintain accurate records, policy documents, lineage reports, and audit findings while ensuring transparency and traceability.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Develop Data Governance Framework and Policies	2	3	-	1
PC2. Communicate and Advocate Data Governance Policies	2	3	-	1
PC3. Define and enforce data standards including data definitions, classifications, quality benchmarks, and security protocols.	2	3	-	1
PC4. Establish metadata management practices to track data lineage, sources, and usage.	2	3	-	1
PC5. Employ tools and processes to monitor adherence to governance policies and standards	2	3	-	1
PC6. Conduct periodic reviews, audits, and risk assessments to detect non-compliance or data breaches.	2	4	-	1
PC7. Implement policies to safeguard sensitive and critical data, ensuring compliance with relevant regulations (e.g., GDPR, HIPAA).	3	4	-	1
PC8. Coordinate with IT and legal teams to enforce data access controls, encryption, and anonymization techniques.	3	4	-	2
PC9. Deploy and manage tools that facilitate policy enforcement, data cataloging, and data lineage tracking.	2	4	-	2
PC10. Leverage automation solutions for policy enforcement and monitoring	2	4	-	2
PC11. Establish procedures for handling data quality issues, policy breaches, or non-compliance incidents.	2	4	-	2
PC12. Lead corrective actions and communicate findings and resolutions to stakeholders.	2	4	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Foster a Culture of Data Governance and Ownership	2	4	-	2
PC14. Maintain Documentation and Records	2	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8165
NOS Name	Implement and Enforce Data Governance Policies
Sector	IT-ITeS
Sub-Sector	
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8101.Extract or Import data as per requirement	30	50	-	20	100	17
SSC/N8102.Process the data	30	50	-	20	100	17
SSC/N8163.Ensure organizational quality standards are achieved regarding data	30	50	-	20	100	17
SSC/N8164.Application of new and emerging tools	30	50	-	20	100	17
SSC/N8165.Implement and Enforce Data Governance Policies	30	50	-	20	100	17
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	15
Total	170	280	-	100	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.